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INTERIM REVISION OF
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INTERIM FEDERAL SPECIFICATION

CHEMICALS, PHOTOGRAPHIC "SINGLE SUBSTANCE",
GENERAL SPECIFICATION FOR

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Electrical and Mechanical Division (FME), Washington, D. C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers requirements and tests for chemicals used in photographic processing.

1.2 Classification.

1.2.1 Grade. This specification covers one grade of photographic chemicals designed as "photographic grade," (see 6.3).

1.2.2 Classes. The photographic chemicals covered by this specification shall be of the following classes:

Column 1	Column 2
	ANSI Standards (See 2.2)
Class I - Acids	
Acetic Acid, Glacial	PH4.100
Sulfuric Acid	PH4.101
Citric Acid, Monohydrate	PH4.102
Boric Acid, Crystalline	PH4.103
Hydrochloric Acid	PH4.104
Sodium Acid Sulfate, Fused	PH4.105
Acetic Acid, 28-Percent Solution	PH4.106
Citric Acid, Anhydrous	PH4.107
Class II - Developing agents	
p-Metylaminophenol Sulfate (Armol, Elon, Genol, Graphol, Metol, Photol, Pictol, Rhodol, Veritol, Verol)	PH4.125
Hydroquinone (Para-Dihydroxybenzene, Quinol, Hydroquinol)	PH4.126
2, 4-Diaminophenol Hydrochloride (Acrol, Amidol, Dianol, Dolmi)	PH4.127
p- Hydroxyphenylglycine (Athenon, Glycin, p-Hydroxyphenylaminoacetic Acid, Iconyl, Monazol)	PH4.128
p-Aminophenol Hydrochloride (Kodelon, P.A.P.)	PH4.129

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Column 1	Column 2
	ANSI Standards (See 2.2)
Class II - Developing agents	
Chlorohydroquinone (Adurol, Chloroquinol, CHQ, Monochlorohydroquinone, 2-Chloro-1, 4-Dihydroxybenzene)	PH4.134
p-Benzyl-Aminophenol Hydrochloride (Duratol, Mono-Benzyl-para-Aminophenol Hydrochloride, N-Benzyl-p-Aminophenol Hydrochloride)	PH4.135
1-Phenyl-3 Pyrazolidone (Phenidone, Phenidone A, Graphidone)	PH4.136
Class III - Hardeners	
Aluminum Potassium Sulfate	PH4.150
Chromium Potassium Sulfate	PH4.151
Formaldehyde, 37-Percent Solution (Formalin)	PH4.152
Paraformaldehyde (Trioxymethylene)	PH4.153
Aluminum Chloride Solution	PH4.154
Sodium Formaldehyde Bisulfite, Anhydrous	PH4.156
Class IV - Miscellaneous	
Sodium Sulfate, Anhydrous	PH4.175
Sodium Acetate, Anhydrous	PH4.176
Sodium Thiocyanate	PH4.177
Isopropylamine, 50-Percent Aqueous Solution (Monoisopropylamine)	PH4.178
Sodium Citrate, Dihydrate	PH4.179
Benzyl Alcohol	PH4.181
Ammonium Chloride	PH4.183
Ammonium Sulfate	PH4.184
Ethylenediaminetetraacetic Acid and its Salts (EDTA)	PH4.185
Hydroxylamine Sulfate (Hydroxylammonium Sulfate) (Oxammonium Sulfate)	PH4.186
Hydroxylamine Hydrochloride (Hydroxylammonium Chloride)	PH4.187
Ethylenediamine	PH4.188
Class V - Restrainers and Antifoggants	
A. Inorganic	
Potassium Bromide	PH4.200
Potassium Iodide	PH4.201
Potassium Chloride	PH4.202
Sodium Chloride	PH4.203
Sodium Bromide	PH4.207
B. Organic	
Benzotriazole (1, 2, 3-Benzotriazole)	PH4.204
5-Methylbenzotriazole	PH4.205
5-Nitrobenzimidazole Nitrate	PH4.206

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Column 1	Column 2 ANSI Standards (See 2.2)
Class VI - Alkalies	
Sodium Hydroxide	PH4.225
Potassium Hydroxide	PH4.226
Sodium Carbonate, Monohydrate	PH4.227
Sodium Carbonate, Anhydrous	PH4.228
Potassium Carbonate, Anhydrous and Sesquihydrate	PH4.229
Sodium Tetraborate, Pentahydrate and Decahydrate (Borax, 5 Mole and Borax, 10 Mole)	PH4.230
Sodium Metaborate, Octahydrate	PH4.231
Ammonium Hydroxide	PH4.232
Trisodium Phosphate, Dodecahydrate	PH4.234
Class VII - Fixing Agents	
Sodium Thiosulfate, Anhydrous and Sodium Thiosulfate, Crystalline (Anhydrous Hypo and Crystal Hypo)	PH4.250
Ammonium Thiosulfate Solution (60-Percent Solution) (Ammonium Hypo Solution)	PH4.252
Class VIII - Sulfites	
Sodium Sulfite, Anhydrous	PH4.275
Sodium Bisulfite, Anhydrous	PH4.276
Potassium Metabisulfite	PH4.277
Class IX - Bleaching Agents	
Potassium Dichromate (Potassium Bichromate)	PH4.300
Potassium Permanganate	PH4.301
Potassium Ferricyanide	PH4.302
Potassium Persulfate	PH4.303
Sodium Ferrocyanide	PH4.304

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-C-300 - Chemicals, Liquid, Packaging and Packing of.
PPP-P-16 - Packaging of Photographic Dry and Liquid Chemicals.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specification and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

National Motor Freight Traffic Association, Incorporated Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

Code of Federal Regulations:

Title 21, Food and Drugs, Chapter 1 Food and Drug Administration (21 CFR 170-299).

Part 299 - Regulations under the Federal Caustic Poison Act.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.)

American National Standards Institute, Inc. Standards listed in 1.2.2.

(Copies may be obtained from the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Requirements of chemicals listed in column 1, paragraph 1.2.2, covered by this specification shall conform to the requirements of the American National Standards Institute Standards listed in column 2, paragraph 1.2.2.

3.2 Workmanship. Workmanship shall be in accordance with manufacturing techniques and practices covering manufacture of photographic chemicals as indicated in the applicable American National Standards Institute Standards.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Manufacturer. The manufacturer shall perform the sampling and inspection specified herein prior to submission of the product to the Government for acceptance.

4.2.1 Test equipment and inspection facilities. The manufacturer shall furnish and maintain all test equipment and inspection facilities for performing all inspection, and shall maintain complete inspection records. Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of the required inspection. The manufacturer shall establish the accuracy and provide for the calibration of test equipment to the satisfaction of the Government. Manufacturers not having test equipment and inspection facilities satisfactory to the Government will engage the services of a commercial laboratory acceptable to the Government.

4.2.2 Inspection and test data. Records of control tests made to determine that each batch of a given photographic single substance chemical has been accurately formulated shall be kept by the contractor and made available to the Government.

4.3 Government. Acceptance of any chemical product covered by this specification shall be based upon verification by the Government of the manufacturer's compliance with the requirements, and sampling and inspection procedures of this specification. The Government may, at its option, repeat any or all of the inspections specified herein and in applicable referenced specifications or standards of the American National Standards Institute.

4.4 Sampling. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.4.1 Batch. A batch will depend on the method of manufacture. For continuous modes of manufacture, a batch shall be defined as the entire quantity produced during one production cycle or one eight-hour period, whichever occurs first. For other than continuous modes, a batch shall consist of that amount of finished chemical resulting from one reaction mixture "charge".

4.4.2 Lot. For the purpose of sampling, a lot shall consist of all the chemicals of the same type, manufactured as one "batch" and offered for acceptance at one time.

4.4.3 Samples from lots. At the discretion of the procuring activity, samples of any or all lots of chemicals being supplied under contract shall be furnished without additional cost to the Government. These samples, consisting of one or more unopened containers, shall be selected at random from the material being supplied under contract, by the contractor, or by the Government inspector or authorized representative of the procuring activity, whoever is specified. These samples will be used in performing verification inspection. Samples consisting of unit packages, shall be selected at predetermined intervals, to assure sampling from the start, the finish, and the midway point, when appropriate, of each production run.

4.4.3.1 Sampling for inspection of filled container. A random sample of filled containers shall be selected from each lot in accordance with MIL-STD-105 at inspection level I, and at acceptable quality level (AQL) 2.5 percent defective, to verify compliance with all stipulations of this specification regarding fill, closure, marking, and other requirements not involving tests.

4.4.3.2 Sampling for tests. When practicable, for purposes of analysis, at least one original unopened container shall be taken at random from each lot segregated for inspection. When this is not practicable, a representative composite sample shall be taken, placed immediately in a clean dry glass bottle, sealed, marked, and submitted for testing, as specified in 4.5.4. The sample shall be not less than two pounds nor more than 100 pounds.

4.5 Quality conformance inspection.

4.5.1 Inspection conditions and methods. Unless otherwise specified in the required test of the applicable referenced specification or standard of the American National Standards Institute, all inspections shall be made at ambient room temperature, pressure, and humidity.

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4.5.2 Inspection of filled containers. Each sample-filled container selected in accordance with 4.4.3.1 shall be examined for defects of the container and closure, for evidence of leakage, and for unsatisfactory markings. Each sample-filled container shall also be weighed to determine the amount of the contents. Any container in the sample having one or more defects, or under required fill, shall be rejected.

4.5.3 Inspection of preparation for delivery. An inspection shall be made to determine that the packaging and packing and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table I. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 defects per hundred units.

TABLE I. Classification of preparation for delivery defects

Examine	Defects
Marking (exterior and interior)	Omitted; incorrect; illegible, improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.
Contents (exterior and interior)	Number per container is more or less than required. Net weight exceeds requirements.

4.5.4 Inspection tests. Samples selected in accordance with 4.4.3.2 shall be subjected to the tests specified in the applicable standard of the American National Standards Institute (see 1.2.2, column 2.) If any sample fails in one or more of the applicable tests, the entire lot represented by the sample shall be rejected.

4.5.5 Rejected lots. Disposition of defective units or rejected lots shall be in accordance with MIL-STD-105.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Photographic chemicals in quantities, as specified (see 6.2), shall be packaged as specified for level A in PPP-P-16 and PPP-C-300 as appropriate.

5.1.2 Level C. Photographic chemicals shall be packaged in uniform quantities with sufficient outage provided in unit containers to prevent leakage or distortion due to expansion of contents through temperature changes occuring in transit from supplier to initial destination.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Photographic chemicals shall be packed in accordance with the requirements of level A of PPP-P-16 and PPP-C-300 as appropriate.

5.2.2 Level B. Photographic chemicals shall be packed in accordance with the requirements of level B of PPP-P-16 and PPP-C-300 as appropriate.

5.2.3 Level C. Photographic chemicals packaged as specified in 5.1, shall be packed in quantities, as specified (see 6.2), in containers that will assure safe arrival at destination in compliance with the National Motor Freight Classification Rules or the Uniform Classification Rules, as applicable.

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5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to markings required by contract or order, the interior and the shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The chemicals covered by this specification are intended for use in photographic processing.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, symbol, and date of this specification.

(b) Total weight required and size or weight of individual containers.

(c) Selection of the level of packaging and packing required (see 5.1 and 5.2).

(d) Marking, if other than specified (see 5.3).

(e) Responsibility for inspection, if other than specified (see 4.1).

(f) Technical literature required.

(g) Additional marking in accordance with the Federal Caustic Poison Act, as required.

(h) Manufacturer's name and batch number.

6.3 Photographic grade. The term "photographic grade" refers to chemicals in which impurities known to be photographically harmful are limited to a safe quantity, while inert impurities are restricted to amounts not reducing the required assay strength.

MILITARY CUSTODIANS:

Army-EL
Navy-AS
Air Force-70

Review activities:

Navy-AS
Air Force-70
Other-NSA, GS

User activity:

Army-AT

Preparing activity:

GSA-PSS

GSA DC 74-10770

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